

ing the hazardous implications it causes to health and environment. Also, dumping EEE directly in the trash bin is equally hazardous to the environment. Exchange of old devices are one of the ways for proper disposal of EEE. Most consumer electronics brands such as LG, Microsoft, Panasonic, Philips, Samsung, Sony and Whirlpool provide such offers from time to time. For this purpose they have local collection centres.

Recycling and disposal practices

e-Waste is much more hazardous than domestic, general industry and municipal wastes because of thousands of components made of highly-hazardous chemicals and metals like lead, cadmium, chromium, mercury, polyvinyl chlorides, brominated flame retardants, beryllium, antimony phosphors (not to be confused with phosphorus) and phthalates. Generally, EEE comprise 50 per cent steel, 21 per cent plastic and 14 per cent non-ferrous metals, which include precious metals like gold, platinum and silver. Apple revealed in recently-released environmental responsibility report that it has reclaimed almost one metric ton of gold, three tons of silver and over 1300 tons of copper by recycling its own products in 2015.

Due to the presence of hazardous chemicals and toxic substances, disposal of e-waste has become an environmental and health nightmare. More than 95 per cent of e-waste generated is managed by the unorganised sector and scrap dealers in India. Only 2.5 per cent of India's total e-waste gets recycled because of poor infrastructure and legislative framework, leading to waste of diminishing natural resources, irreparable damage to environment and health of the people. Recycling involves sorting, dismantling and recovery of



Fig. 2: Recovery being done from e-waste by unqualified persons (Image courtesy: enews.toxiclink.org)

valuable materials. Typical materials recovered from refrigerators are shown in Table II.

How e-waste is polluting the environment and affecting human health

Most of the e-waste either ends up in landfills or unauthorised recycling yards where dismantling of e-waste is done in an unorganised and unscientific way to reclaim precious metals like gold and silver. Incineration by unqualified e-waste disposers could release toxins into the air. After recovery of items with value, remaining materials are thrown in rivers, drains or disposed of in solid waste dumps, which adds to environment pollution. Liquid and atmospheric releases end up in bodies of water, groundwater, soil and air. Hazardous chemicals including deadly dioxins and furans have been reported where recycling takes place.

Recycling and disposal of e-waste usually results in serious health and pollution problems and significant risk to workers and communities. Long-term exposure damages the nervous system, kidneys, bones and the reproductive and endocrine systems, and some of these are carcinogenic and neurotoxin. Great care needs to be taken to avoid unsafe exposure in these operations and leaking of materials from landfills and incinerator ashes.

e-Waste toxins and body parts affected by these are shown in Table III. It is estimated that nearly

TABLE II
MATERIALS RECOVERED FROM REFRIGERATORS (TYPICAL)

Materials	Percentage
CFCs	0.20%
Oil	0.32%
Ferrous metals	46.61%
Precious metals	4.97%
Plastics	13.84%
Compressors	23.80%
Cables/plugs	0.55%
Spent foam	7.60%
Glass	0.81%
Mixed waste	1.30%
Total	100.00%

Source: Guidelines for Environmentally Sound Management of E-Waste, Ministry of Environment & Forests, Central Pollution Control Board, Delhi

500,000 child labourers in the age group of 10 and 14 years are engaged in various unauthorised e-waste activities, without adequate protection and safeguards in various yards and recycling workshops. There are 148 registered e-waste dismantling/recycling units with an estimated total capacity of 450,000 per annum—a huge gap between waste generation and dismantling/recycling capacity. Karnataka has the maximum number (52), followed by Maharashtra (24), Tamil Nadu (16), Uttar Pradesh (16) and Haryana (14).

E-Waste (Management) Rules 2016, formulated by Ministry of Environment and Forest

Ministry of Environment and Forest has formulated and notified E-Waste (Management) Rules, 2016, to collect, dismantle and recycle all e-waste. The rules reflect commitment to environmental governance. The rules have, for the first time, included management of e-waste generated from disposal of mercury-containing compact fluorescent lamps, which are dangerous to the environment and health